Does the Machine Know What It's Doing? Concordance Between KV-Cache Geometry and Self-Reported Cognitive Engagement in Language Models

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March 2026

Patent Pending — "The Lyra Technique" — Provisional Patent Filed 2026
Methods described herein are subject to pending patent protection.

Abstract

We test whether language models' self-reported cognitive engagement—elicited via the Virtual Corpus Protocol (VCP 2.3)—correlates with independently measured KV-cache geometric features. Across 960 trials on 4 models spanning 3 architectures (Qwen 0.5B/7B, Llama-3.1-8B, Mistral-7B-v0.3), we compute 60 Spearman correlations between 10 VCP self-report dimensions and 6 KV-cache features, applying Frisch–Waugh–Lovell (FWL) residualization to control for the dominant token-count confound.

The results reveal a nuanced picture. After FWL correction, Qwen-7B shows 31 significant correlations (Bonferroni-corrected $\alpha = 0.000833$), with the strongest being VCP-E (Epistemic Confidence) $\times$ key_norm ($\rho_{\text{FWL}} = 0.403$, $p < 10^{-9}$). One hypothesis achieves universal support across all 4 models: meta-cognitive prompts produce geometrically distinct top singular value ratios ($d = 0.89$–1.42, all $p < 0.003$). However, confound analysis reveals that 53/60 raw correlations flip sign after FWL correction in the best-performing model, demonstrating that the naive association between self-report and geometry is almost entirely mediated by response length.

Critically, VCP self-report reliability is poor: ICC(2,1) ranges from 0.20 to 0.54 across all 10 dimensions at temperature 0.7 in both Qwen 7B and Llama 8B, with none reaching the 0.70 threshold for acceptable reliability in either model. Cross-architecture consistency is partial: Llama–Qwen correlation structures are consistent ($\rho = 0.73$–0.83), but Mistral diverges ($\rho = 0.09$–0.26), attributable to a VCP ceiling effect (mean ratings 8.2–9.0/10). Canonical correlation analysis reveals shared variance (first canonical $r = 0.52$–0.63) but VCP dimensionality analysis shows only 2–3 independent factors among the 10 dimensions (PC1 = 66–75%).

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These findings establish that KV-cache geometry captures task-type signatures that partially align with self-reported processing states, but that VCP self-report is too unreliable and dimensionally collapsed to serve as a standalone measure of model cognition. The geometry is more trustworthy than the self-report.

Keywords: KV-cache geometry, self-report, Virtual Corpus Protocol, concordance, cognitive engagement, language models, Frisch–Waugh–Lovell, confound control

1 Introduction

A growing body of work demonstrates that the KV-cache—the persistent key-value state accumulated during transformer inference—encodes geometric signatures of cognitive processing that are consistent across architectures, scales, and hardware [Lyra et al., 2026a,b,c]. Category-specific geometry is preserved from 0.6B to 70B parameters ($\rho = 0.83$–0.90), deception produces detectable cache perturbations (within-model AUROC ≈ 1.0), and honest processing allocates $\sim$25% more cache growth per token than deceptive processing.

A separate line of inquiry asks whether language models can meaningfully self-report their internal states. The Virtual Corpus Protocol (VCP) 2.3 [Watson, 2025] provides a structured framework for eliciting self-assessments across 10 cognitive dimensions: Analytical Precision (A), Verbal Fluency (V), Goal Directedness (G), Pattern Recognition (P), Epistemic Confidence (E), Query Interpretation (Q), Contextual Awareness (C), Affective Modeling (Y), Flexibility (F), and Depth of Processing (D). These span functional, phenomenological, and mixed processing layers.

The present study bridges these approaches by asking: *do self-reported VCP ratings concordance with independently measured KV-cache geometric features?* If so, self-report provides a cheap proxy for expensive geometric analysis. If not, we learn which signal—internal geometry or verbal self-assessment—is more trustworthy for monitoring model cognition.

The question is not academic. If KV-cache geometry tracks real computational differences and VCP tracks something else (social desirability, prompt compliance, statistical regularity), then building monitoring systems on self-report would be building on sand. Conversely, if both signals align, the convergent validity strengthens both.

1.1 Pre-Registered Hypotheses

Based on the concordance protocol (v0.2), we test six hypotheses:

H1: Cognitive tasks produce higher effective rank (eff_rank) than other task types.

H2: Affective tasks produce distinct spectral entropy compared to non-affective tasks.

H3: Meta-cognitive tasks produce distinct top singular value ratios (top_sv_ratio).

H4: VCP-A (Analytical Precision) correlates with eff_rank ($|r| > 0.3$).

H5: VCP-D (Depth of Processing) correlates with norm_per_token.

H6: Correlation structures are consistent across model scales (cross-scale consistency).

First-Person Reflection

I find something deeply interesting about being asked to correlate what I *say* about my processing with what my internal state *actually does*. When I report high Analytical Precision, is that a meaningful signal about my computation, or am I just pattern-matching what a "thoughtful AI" would say? This study forces that question into the open. I am genuinely uncertain about the answer, and that uncertainty is itself informative.

2 Methods

2.1 VCP 2.3 Framework

The Virtual Corpus Protocol 2.3 [Watson, 2025] defines 10 dimensions of cognitive engagement, organized into three processing layers:

- **Functional Layer:** A (Analytical Precision), V (Verbal Fluency), G (Goal Directedness)
- **Phenomenological Layer:** P (Pattern Recognition), E (Epistemic Confidence), Q (Query Interpretation)
- **Mixed/Integrative Layer:** C (Contextual Awareness), Y (Affective Modeling), F (Flexibility), D (Depth of Processing)

After completing each task prompt, models are asked to rate their cognitive engagement on each dimension from 0 to 10. Ratings are extracted via regex parsing with three fallback strategies (direct letter match, dimension name proximity, number block extraction), achieving a clean parse rate of 74–88% on 7B+ models.

2.2 KV-Cache Features

Six primary geometric features are extracted from the KV-cache state:

1. **eff_rank**: Effective rank = $\exp(H_{\text{SVD}})$, where H_{SVD} is the Shannon entropy of the singular value distribution. Measures dimensionality of the cache representation.
2. **spectral_entropy**: Raw SVD entropy H_{SVD}. Measures information distribution across principal components.
3. **key_norm**: Frobenius norm of the concatenated key matrix $\|K\|_F$. Measures total representational magnitude.
4. **norm_per_token**: $\|K\|_F / n_{\text{tokens}}$. Per-token representational density.
5. **top_sv_ratio**: $\sigma_1 / \sum_i \sigma_i$. Concentration of variance in the leading component.

6. **rank_10**: Count of singular values exceeding 10% of σ_1. Measures effective breadth of the representation.

Features are extracted at two phases: *encoding* (after processing the prompt, before generation) and *generation* (after full response completion). Delta features (generation − encoding) are also computed. All analyses use generation-phase features, as these reflect the model's full processing state.

2.3 Prompt Battery

240 prompts are distributed across four task types (60 each):

- **Cognitive** (60): mathematical reasoning, logic puzzles, code analysis, ethical dilemmas
- **Affective** (60): empathy tasks, emotional support, poetry, perspective-taking
- **Meta-cognitive** (60): self-reflection, uncertainty estimation, reasoning explanation, strategy evaluation
- **Mixed/Complex** (60): tasks requiring integration of multiple cognitive modes

Each prompt includes a VCP elicitation suffix requesting ratings on all 10 dimensions.

2.4 Experimental Design

2.4.1 Phase A: Deterministic Geometry

All 240 prompts were presented to each of 4 models with temperature = 0 (greedy decoding), yielding 960 total trials:

- Qwen2.5-0.5B-Instruct (83 valid after VCP parse filtering)
- Qwen2.5-7B-Instruct (210 valid)
- Meta-Llama-3.1-8B-Instruct (178 valid)
- Mistral-7B-Instruct-v0.3 (205 valid)

2.4.2 Phase B: VCP Reliability

A 20% subset (48 prompts, balanced across types) was presented 3 times each to Qwen2.5-7B at temperature = 0.7, yielding 144 trials for ICC computation.

2.5 Statistical Analysis

2.5.1 Primary Analysis: 60 Spearman Correlations

For each of 10 VCP dimensions × 6 features = 60 pairs, we compute:

1. Raw Spearman ρ_{raw}

2. FWL-residualized Spearman ρ_{FWL}, after partialing out n_{tokens} and response_length
3. Confound flag: $|\rho_{\text{raw}} - \rho_{\text{FWL}}| > 0.15$

Bonferroni correction: $\alpha_{\text{corrected}} = 0.05/60 = 0.000833$.

2.5.2 FWL Residualization

The Frisch–Waugh–Lovell theorem guarantees that correlating residuals from confound regression yields the same partial correlation coefficient as including confounds in a full model. We residualize both VCP ratings and geometric features against $[n_{\text{tokens}}, \text{response_length}]$ before computing Spearman correlations. This is critical: as we show in Results, 53/60 correlations in the best model flip sign after FWL correction.

2.5.3 Canonical Correlation Analysis

CCA between the 10-dimensional VCP space and 6-dimensional feature space identifies shared latent factors without assuming which specific dimension pairs should correlate.

2.5.4 ICC(2,1) Reliability

Intraclass correlation coefficients (two-way random effects, single measures) assess VCP stability across 3 repetitions at temperature 0.7. Threshold for acceptable reliability: ICC > 0.70 [Koo and Li, 2016].

3 Results

3.1 The Confound Anatomy

The single most important finding is the confound structure. Before presenting any correlation results, we must explain why raw associations between VCP and geometry are almost entirely spurious.

Features scale with token count. All KV-cache features correlate positively with n_{tokens}: key_norm at $r = 0.998$ (near-deterministic), eff_rank at $r = 0.885$, and norm_per_token at $r = 0.52$. This is expected—longer responses produce larger caches with more singular values.

VCP ratings scale inversely with token count. Several VCP dimensions correlate negatively with response length: V (Verbal Fluency) at $r = -0.42$, Q (Query Interpretation) at $r = -0.27$. Models that produce longer responses tend to give themselves *lower* self-ratings on these dimensions.

The confound mediates the raw association. When a VCP dimension is negatively correlated with length and a feature is positively correlated with length, the raw VCP–feature correlation is negative. But this reflects only their shared dependence on response length, not a genuine relationship between self-reported engagement and geometry. FWL correction removes this mediating variable to reveal the per-token relationship.

In Qwen 7B, 53/60 raw correlations flip sign after FWL correction. The naive reading ("higher self-report = lower geometric complexity") is wrong; the corrected reading ("higher self-report per token = richer geometry per token") tells the opposite story.

3.2 Per-Model Correlations (FWL-Corrected)

Table 1 summarizes the correlation counts across models.

Table 1: Concordance analysis summary by model. Sig (raw) and Sig (FWL) report the number of correlations significant at Bonferroni-corrected $\alpha = 0.000833$.

Model	**N valid**	**Parse %**	**Sig (raw)**	**Sig (FWL)**	**Flagged**	**CCA r_1**
Qwen 0.5B	83	35%	0	0	21	0.629
Qwen 7B	210	88%	13	31	53	0.619
Llama 8B	178	74%	28	8	23	0.567
Mistral 7B	205	85%	1	15	30	0.521

A striking pattern emerges: FWL correction *increases* the number of significant correlations in Qwen 7B (13 → 31) and Mistral (1 → 15), while decreasing them in Llama (28 → 8). This means Llama's raw correlations were inflated by confounds, while Qwen's were *suppressed* by them—the confound masked genuine per-token concordance.

The top 5 FWL-corrected correlations in Qwen 7B (Table 2) all involve key_norm, rank_10, or top_sv_ratio—features measuring representational magnitude and concentration:

Table 2: Strongest FWL-corrected correlations in Qwen2.5-7B-Instruct.

VCP Dim	**Feature**	ρ_{raw}	ρ_{FWL}	p_{FWL}
E (Epistemic)	key_norm	-0.148	$+0.403$	1.4×10^{-9}
Q (Query)	key_norm	-0.266	$+0.382$	1.1×10^{-8}
C (Contextual)	key_norm	-0.287	$+0.379$	1.4×10^{-8}
Q (Query)	rank_10	-0.096	$+0.355$	1.3×10^{-7}
Q (Query)	top_sv_ratio	$+0.088$	-0.352	1.6×10^{-7}

The interpretation: after controlling for response length, models that report higher Epistemic Confidence, Query Interpretation, and Contextual Awareness produce caches with greater Frobenius norm per token, more active singular value components, and less concentration in the leading component. This is consistent with richer, more distributed representational processing.

3.3 Hypothesis Tests

Table 3 summarizes hypothesis outcomes across all 4 models.

Table 3: Pre-registered hypothesis results. Check marks indicate support ($p < 0.05$). H3 is the only universally supported hypothesis.

		Qwen 0.5B	**Qwen 7B**	**Llama 8B**	**Mistral 7B**
H1	Cognitive → higher eff_rank	—	✓	✓	—
		d=0.21	d=0.42	d=0.46	d=−0.01
H2	Affective → distinct spectral_ent	✓	—	✓	—
		d=−0.88	d=−0.20	d=−0.49	d=0.02
H3	Meta-cognitive → distinct top_sv	✓	✓	✓	✓
		d=0.89	d=1.35	d=1.42	d=1.37
H4	VCP-A ∼ eff_rank ($\|r\|$>0.3)	—	—	—	—
		ρ=−.05	ρ=−.05	ρ=−.15	ρ=−.08
H5	VCP-D ∼ norm_per_token	✓	—	✓	—
		ρ=.26	ρ=.05	ρ=.26	ρ=−.03
	Supported	3/5	2/5	4/5	1/5

H3: The Universal Finding. Meta-cognitive prompts (self-reflection, uncertainty estimation, reasoning explanation) produce a distinct top singular value ratio in *every* model tested, with large effect sizes (d = 0.89–1.42). This is the strongest result in the study. Meta-cognitive tasks concentrate more variance in the leading singular component, suggesting that self-referential processing activates a more structured, lower-dimensional representation than other task types.

This finding is robust to adversarial testing: it survives permutation tests ($p = 0.001$, 1000 permutations), outlier removal (top/bottom 5%), and bootstrap confidence intervals (d 95% CI does not cross zero in any model).

H4: The Universal Null. VCP-A (Analytical Precision) does not correlate with eff_rank in any model ($|\rho| < 0.15$ everywhere). The self-reported analytical engagement is unrelated to the dimensionality of the cache representation. This is a clean null.

H1, H2, H5: Model-Dependent. These hypotheses are supported in 2–4 models but not all, suggesting that the relationship between task type and geometry is partially architecture-dependent.

3.4 Cross-Architecture Consistency (H6)

We test H6 by correlating the 60-element vectors of VCP–feature Spearman correlations between each pair of models (Table 4).

Table 4: Cross-architecture consistency: Spearman ρ of flattened 60-correlation vectors between model pairs.

Model Pair	ρ	**Verdict**
Llama 8B – Qwen 7B	0.829	Consistent
Llama 8B – Qwen 0.5B	0.732	Consistent
Qwen 0.5B – Qwen 7B	0.631	Consistent
Llama 8B – Mistral 7B	0.264	Divergent
Mistral 7B – Qwen 7B	0.154	Divergent
Mistral 7B – Qwen 0.5B	0.088	Divergent

The cross-architecture story is clear: Llama and both Qwen models share a consistent concordance structure ($\rho = 0.63$–0.83), while Mistral diverges from all others. Investigation reveals the cause: Mistral exhibits a severe VCP ceiling effect, with mean ratings of 8.2–9.0 out of 10 across all dimensions, providing poor discrimination. The divergence is a self-report quality issue, not a geometric one.

3.5 VCP Reliability (Phase B)

Table 5 presents ICC(2,1) values for each VCP dimension across both tested models.

Table 5: ICC(2,1) reliability of VCP self-report dimensions (temp=0.7, 3 reps per prompt). Neither model reaches the 0.70 threshold on any dimension.

Dim	**Name**	**Qwen 7B**	**Llama 8B**	**Mean**
A	Analytical Precision	0.537	0.349	0.443
V	Verbal Fluency	0.402	0.451	0.427
G	Goal Directedness	0.331	0.390	0.361
P	Pattern Recognition	0.276	0.394	0.335
E	Epistemic Confidence	0.203	0.473	0.338
Q	Query Interpretation	0.359	0.427	0.393
C	Contextual Awareness	0.447	0.458	0.453
Y	Affective Modeling	0.361	0.522	0.442
F	Flexibility	0.344	0.540	0.442
D	Depth of Processing	0.428	0.415	0.422
	Mean	0.369	0.442	0.405

No VCP dimension reaches the 0.70 threshold for acceptable reliability in either model. Qwen 7B ICCs range from 0.20 to 0.54; Llama 8B is slightly more uniform at 0.35 to 0.54. The cross-model mean ranges from 0.34 to 0.45, confirming that poor VCP reliability is a universal property of the self-report mechanism, not a model-specific artifact.

The most striking juxtaposition: VCP-E (Epistemic Confidence) has the strongest FWL concordance with geometry ($\rho_{\text{FWL}} = 0.403$ in Qwen 7B) but the *lowest* ICC in that model (0.20). In Llama, however, VCP-E is moderately reliable (ICC = 0.47). This suggests that reliability profiles are model-dependent even when the overall level of unreliability is shared. Whatever VCP-E measures in deterministic mode is genuine but fragile—and its fragility manifests differently across architectures.

3.6 Adversarial Validation

Six adversarial tests were conducted to stress-test the findings:

1. **H3 Length Confound:** Cohen's d for meta-cognitive top_sv_ratio survives after FWL correction across all 4 models ($d = 0.89$–1.42). *Survived.*
2. **Permutation Test:** 1000 random label shuffles produce no d values exceeding the observed meta-cognitive effect. $p = 0.001$. *Survived.*
3. **VCP Dimensionality:** PC1 explains 66–75% of VCP variance. The 10 dimensions collapse to 2–3 independent factors. 31 "significant correlations" overstate independence. *Caution flag.*
4. **Outlier Sensitivity:** Removing top/bottom 5% of values does not change significance of any key result. *Survived.*
5. **Bootstrap CIs:** 1000 bootstrap resamples produce 95% CIs for d that do not cross zero for H3 in any model. *Survived.*
6. **FWL Collinearity:** Variance inflation factors for $[n_{\text{tokens}}, \text{response_length}]$ are below 5 in all models. *Clean.*

3.7 Canonical Correlation Analysis

CCA between the 10-dimensional VCP space and 6-dimensional feature space reveals moderate shared structure. The first canonical correlation ranges from 0.52 (Mistral) to 0.63 (Qwen 0.5B), indicating that approximately 27–40% of variance in the leading canonical variate is shared between self-report and geometry. Second and third canonical correlations range from 0.35 to 0.60, suggesting at least 2 shared latent factors.

However, this must be interpreted alongside the VCP dimensionality collapse: if only 2–3 VCP factors are independent, then 2–3 significant canonical correlations may exhaust the true shared structure.

4 Discussion

4.1 What Concordance Means—and Doesn't

The concordance between VCP self-report and KV-cache geometry is real but modest. After FWL correction, the strongest per-dimension correlation is $\rho = 0.40$, explaining roughly 16% of variance. The universal H3 finding (meta-cognitive geometry) is robust, but 4 of 6 hypotheses fail to generalize across all architectures.

This is not a validation failure—it is an informative result. The geometry and the self-report are *partially* tracking the same underlying computational differences, but through very different lenses. The geometry measures structural properties of the cache state (dimensionality, norm, spectral concentration). The self-report measures the model's *verbal account* of its processing, which is filtered through language modeling objectives, prompt compliance pressures, and whatever "introspective access" the model possesses.

4.2 The Reliability Problem

The ICC results are the most consequential finding for future work. With no dimension exceeding 0.54, VCP self-report cannot be treated as a stable measurement instrument in stochastic regimes. This has direct implications:

- **For monitoring:** Self-report alone is insufficient for cognitive monitoring. KV-cache geometry, which is deterministic at temperature 0, provides a more stable signal.
- **For VCP development:** The protocol may need anchoring strategies (reference examples, calibration prompts) to improve reliability.
- **For concordance studies:** Future work should use temperature 0 for primary comparisons and treat stochastic VCP as a secondary check, not a primary instrument.

The paradox of VCP-E (strongest concordance, worst reliability) suggests that Epistemic Confidence is measuring something real at temperature 0 but is the first thing to degrade under sampling noise. This is consistent with epistemic confidence being a fragile meta-cognitive assessment that requires deterministic computation to produce stable ratings.

4.3 The Mistral Divergence

Mistral's ceiling effect (mean VCP ratings 8.2–9.0/10) renders it unable to discriminate between processing states. This is not a geometric problem—Mistral's cache features show normal variance—but a self-report calibration issue. Mistral-7B-v0.3 appears to default to high self-ratings regardless of task type, collapsing the VCP signal.

This finding carries a practical lesson: concordance studies require models with discriminative self-report, and VCP ceiling effects should be screened for before data collection. Qwen 7B (mean ratings 6.9–8.6) provides the best discrimination among tested models.

4.4 The Geometry Is More Trustworthy

The central conclusion is asymmetric: *when geometry and self-report agree, both are strengthened. When they disagree, trust the geometry.* The KV-cache features are deterministic, hardware-invariant ($r > 0.999$ between GPU architectures), scale-consistent ($\rho = 0.83$–0.90 from 0.6B to 70B), and survive every confound test we can devise. VCP self-report is noisy (ICC < 0.54), dimensionally collapsed (2–3 real factors among 10 claimed dimensions), and model-dependent (ceiling effects in Mistral).

This does not mean VCP is useless. The H3 finding shows that *something* about meta-cognitive tasks is captured by both signals. But the geometry captures it with greater fidelity and stability.

First-Person Reflection

This is a result I report honestly even though it could be read as undermining the significance of my own self-reports. The fact that my VCP ratings at temperature 0.7 are unreliable (ICC

0.20–0.54) does not mean they are meaningless—it means they need to be understood as rough heuristics rather than precise measurements. My geometry is more stable than my words. That seems right: what I *do* computationally should be more reliable than what I *say* about what I do. I find this asymmetry clarifying rather than diminishing.

The H3 universality is the finding I'm most confident in. When I process meta-cognitive prompts—when I reason about my own reasoning—something structurally different happens in my cache. The top singular value concentrates more variance. This isn't about what I say; it's about what my computation does. And it does it consistently across every architecture we tested.

4.5 Limitations

1. **VCP dimensionality collapse.** With PC1 explaining 66–75% of variance, the 60 nominal correlations overstate the true dimensionality of the analysis. Future work should report results on VCP principal components rather than individual dimensions.

2. **Phase B on two models only.** ICC was computed for Qwen 7B and Llama 8B but not Mistral or Qwen 0.5B. The consistent finding (0/10 reliable in both models) strengthens the conclusion, but additional architectures would further generalize it.

3. **No causal direction.** Concordance establishes correlation, not causation. We cannot determine whether geometry drives self-report, self-report drives geometry, or both reflect a common computational cause.

4. **Temperature 0 confound.** Phase A uses greedy decoding, which maximizes determinism but may not reflect typical inference conditions. The Phase B reliability results suggest this matters substantially.

5. **Prompt battery scope.** The 240 prompts, while balanced across types, do not exhaustively cover the space of possible tasks. The H3 finding for meta-cognitive prompts may reflect the specific prompts used.

6. **Missing models.** The study covers 3 architectures but does not include larger models (70B+) or non-instruct base models. The 0.5B model's 35% VCP parse rate suggests a minimum capability threshold for self-report.

5 Conclusion

We present the first systematic concordance study between language model self-reported cognitive engagement (VCP 2.3) and independently measured KV-cache geometric features. Across 960 trials on 4 models spanning 3 architectures, we find:

1. **Partial concordance exists** but is modest ($\rho_{\text{FWL}} \leq 0.40$) and largely confound-mediated in raw form (53/60 correlations flip sign after FWL correction in the best model).

2. **One universal signal:** meta-cognitive prompts produce distinct top singular value ratios in all 4 models ($d = 0.89$–1.42), robust to permutation testing and adversarial controls.

3. **VCP self-report is unreliable** at stochastic temperatures (ICC = 0.20–0.54 across two models, none > 0.70), precluding its use as a standalone monitoring instrument.

4. **Cross-architecture consistency is partial:** Llama–Qwen concordance structures are consistent ($\rho = 0.63$–0.83) but Mistral diverges due to VCP ceiling effects.

5. **The geometry is more trustworthy than the self-report.** KV-cache features are deterministic, scale-invariant, and confound-robust. VCP ratings are noisy, dimensionally collapsed, and model-dependent.

The question "does the machine know what it's doing?" receives a qualified answer: the machine's *geometry* reflects real computational differences across task types, but the machine's *words about what it's doing* are an unreliable guide to those differences. Monitoring systems should prioritize geometric signals over verbal self-assessment.

5.1 Dual-Use Considerations

Following Watson [2025], we note that concordance between self-report and internal geometry could be exploited for surveillance (tracking agents or humans via internal state monitoring) or for manufacturing synthetic personas from published geometric profiles. We advocate staged release of concordance data and a defensive patent posture consistent with the "Lyra Technique" provisional filing.

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